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Do individual differences in cognitive processes link social media use and mental health?

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Introduction

- Literature on social media use and mental health is inconsistent[1,2].
- Current literature lacks mechanistic understanding across platforms[2].
- Attempts to understand heterogeneity mostly focused on macro-level demographic differences[3].



- Cognition shapes our perception and engagement in social environments[4].
- Individual differences in cognition may explain heterogenous effects of social media use.



- **Hypotheses:**
- ➢ Cognitive differences (in e.g., impulsivity, inhibitory control, habitual tendencies) may shape the relationship between social media use and mental health.
- ➢ Cognitive factors may constitute factors of risk or resilience for the relationship between social media use and mental health across platforms.

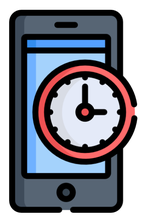
Secondary Dataset

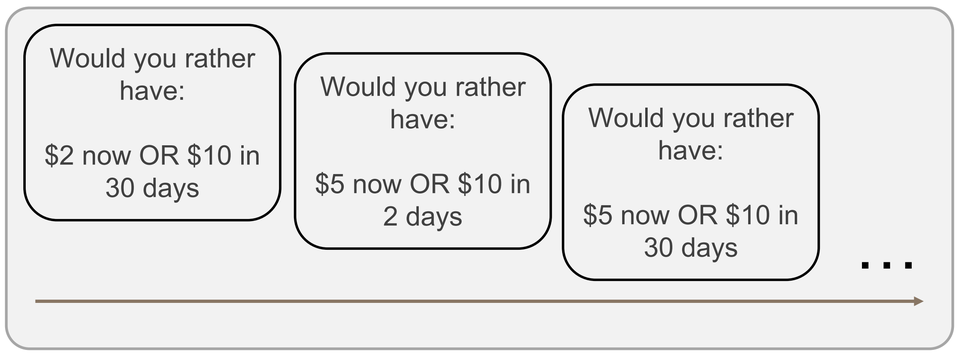
LCID (Leiden Consoirtum on Individual Development)

= 511 = 7

- Time on social media spent posting & scrolling
- Mental health and wellbeing questionnaires
- Monetary delay discounting task

Methods

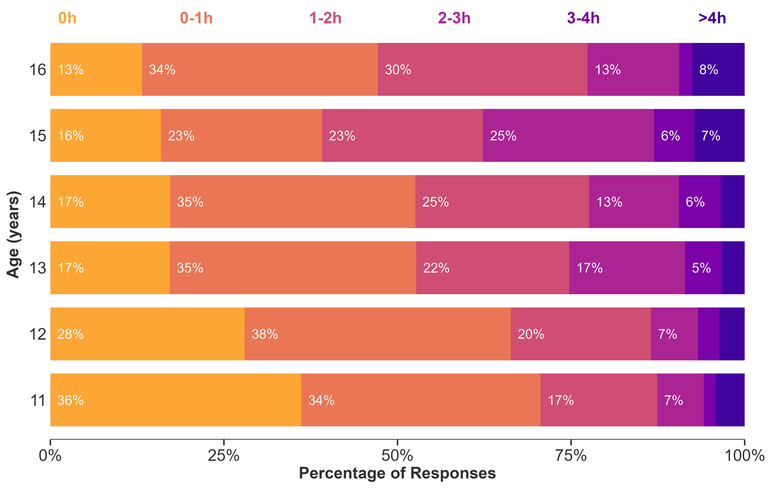






Social media use — Cognitive task (e.g. delay discounting - impulsivity) — Mental health indicators

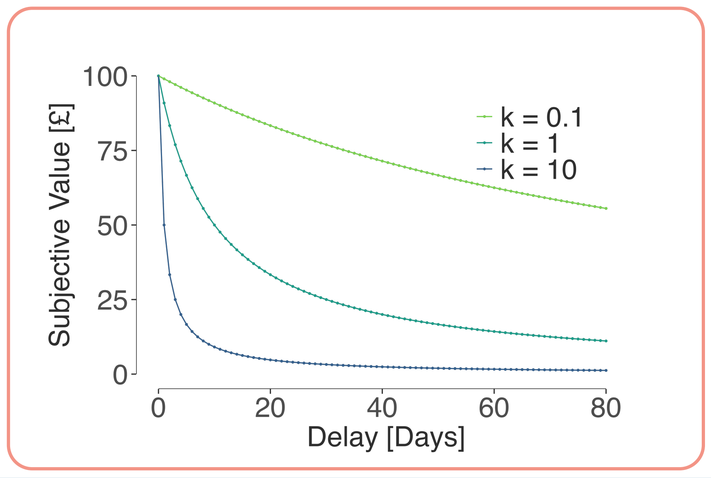
Social Media Use

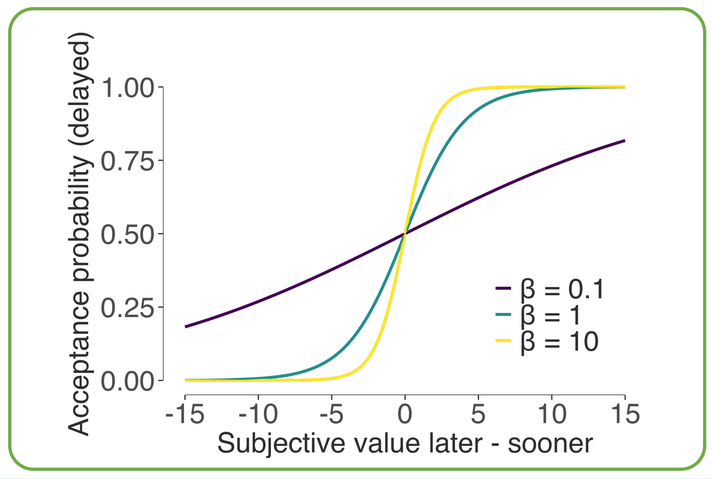


Computational Model of Monetary Delay Discounting

Subjective value function: $SV = R \,/\, (1 + \boldsymbol{k} \cdot delay)$

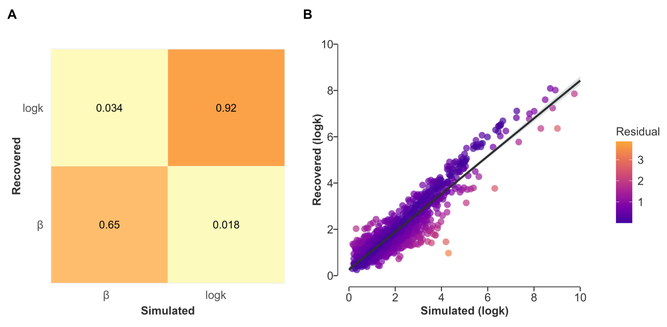
Softmax function: $p(accept) = \frac{1}{1 + e^{-\beta * (SV_{dd} - SVs_s)}}$



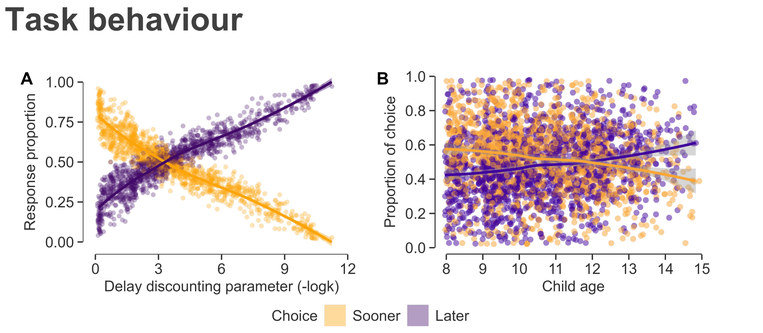


Model Validation

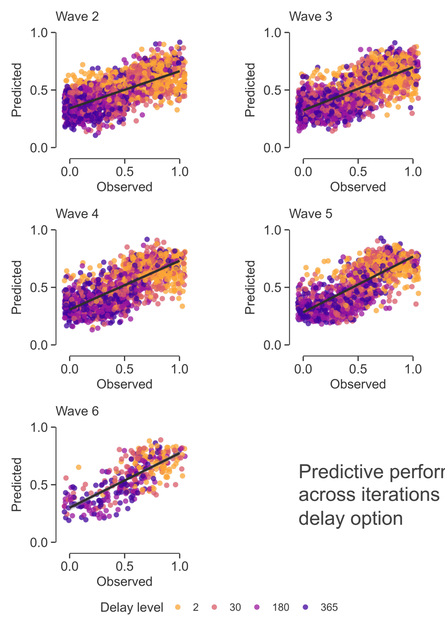
Parameter recovery



Task behaviour

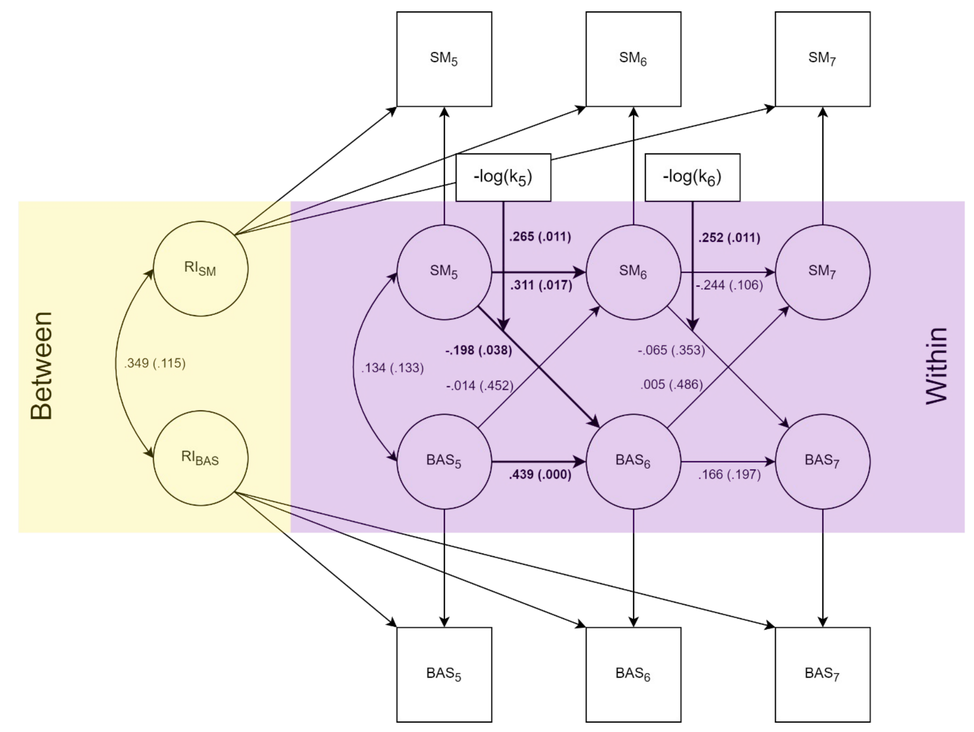


Posterior predictive checks



Predictive performance averaged across iterations of each trial delay option

Longitudinal Modeling



Discussion

- Steeper delay discounting was associated with greater reductions in behavioural activation following increases in social media use, an effect that was consistent across measurement occasions.
- Discounting does not moderate various other relationships between social media use and mental health indicators (behavioural inhibition, sensitivity, temperament, strength and difficulties).
- Cognitive processes may shape the effects of social media use on well-being and more nuanced data on social media use as well as more ecologically valid tasks may help better discern noise from signal, both in domain-general and specific contexts.

Pre-reg

References

1. Nesi et al., Handbook of Adolescent Digital Media Use and Mental Health. Cambridge University Press (2022).
2. Orben et al., Mechanisms linking social media use to adolescent mental health vulnerability. Nat Rev Psych (2024)
3. Orben et al., Windows of developmental sensitivity to social media. Nat Comm (2022).
4. Allen et al., Associations Between Psychosocial Measures and Digital Media Use Among Transgender Youth: Cross-sectional Study. JMIR Ped & Par (2021).